

1. The following data pertain to the resistance (ohms) and failure time (minutes) of certain overloaded resistors.

Resistance (X)	43	29	44	33	33	47	34	31	48	34	46	37
Failure time (Y)	33	20	45	35	22	46	28	29	37	33	47	34

- i) Draw the scatter diagram + comment.
- ii) Calculate the coefficient of correlation r_{xy} .
- iii) Obtain the equations of the two lines of regression.



X	Y	U = X - 33	W = Y - 33	U ²	W ²	U.W
Σ		ΣU	ΣW	ΣU^2	ΣW^2	ΣUW

(ii) Sign of scales are same (scales are + both are +ve)

Hence $r_{xy} = r_{UW} = \frac{n \Sigma UW - (\Sigma U)(\Sigma W)}{\sqrt{n \Sigma U^2 - (\Sigma U)^2} \sqrt{n \Sigma W^2 - (\Sigma W)^2}}$

(iii) $b_{yx} = b_{WU} = \frac{n \Sigma UW - (\Sigma U)(\Sigma W)}{n \Sigma U^2 - (\Sigma U)^2}$

$b_{xy} = b_{UN} = \frac{n \Sigma UW - (\Sigma U)(\Sigma W)}{n \Sigma W^2 - (\Sigma W)^2}$

Calculate $\bar{x}$ and $\bar{y}$

$\bar{x} = 33 + \bar{u}$

$\bar{y} = 33 + \bar{w}$

Eqs of regression lines are

$y = \bar{y} + b_{yx}(x - \bar{x})$

$x = \bar{x} + b_{xy}(y - \bar{y})$

Complete the solution

For the following tri-variate distribution, calculate $R_{1.23}$, $r_{12.3}$ and $r_{23.1}$

X_1	40	46	52	62	44	60	50	54	48	58
X_2	27	33	42	45	27	48	33	39	30	27
X_3	150	100	180	210	170	230	200	240	140	210

Create the table

X_1	X_2	X_3	$U = \frac{X_1 - 52}{2}$	$V = \frac{X_2 - 27}{3}$	$W = \frac{X_3 - 210}{10}$	U^2	V^2	W^2	UV	UW	VW
			ΣU	ΣV	ΣW	ΣU^2	ΣV^2	ΣW^2	ΣUV	ΣUW	ΣVW

$$r_{12} = r_{UV}, \quad r_{13} = r_{UW}, \quad r_{23} = r_{VW}$$

$$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{(1 - r_{23}^2)}$$

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

$$r_{23.1} = \frac{r_{23} - r_{21}r_{31}}{\sqrt{(1 - r_{21}^2)(1 - r_{31}^2)}}$$